

FortiGate Client-to-Site VPN

VPN Client-to-Site con FortiGate y Cliente Linux

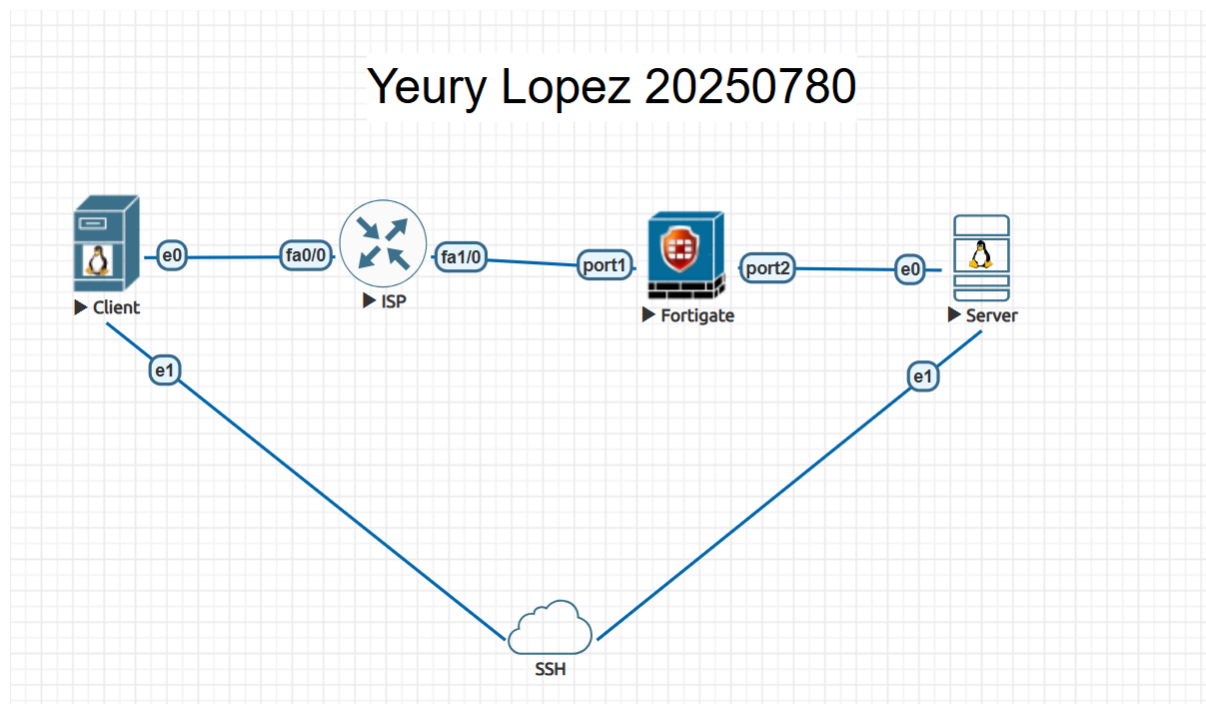
Autor: Yeury Lopez
Matrícula: 20250780
Entregable: P11

1. Objetivo

Configurar una VPN Client-to-Site con FortiGate como servidor VPN y un cliente Ubuntu Linux como usuario remoto. A diferencia de Site-to-Site donde dos redes completas se comunican, en Client-to-Site un usuario individual se conecta remotamente a la red interna. FortiGate asigna una IP del pool VPN al cliente mediante Mode Config, y XAuth autentica al usuario con credenciales de usuario y contraseña. IPSec cifra todo el tráfico entre el cliente y la red interna.

Topología de Red

La topología consiste en un FortiGate como servidor VPN conectado al ISP, un cliente Linux remoto y un servidor Linux en la LAN interna del FortiGate.



Direccionamiento IP

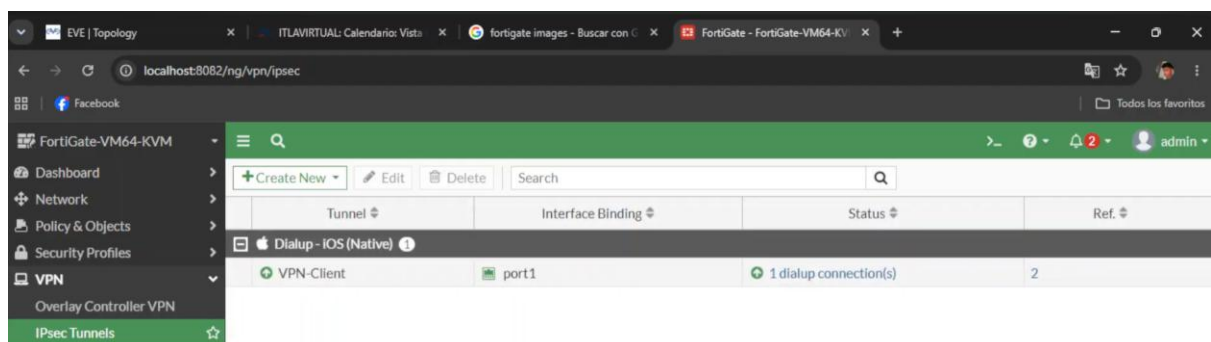
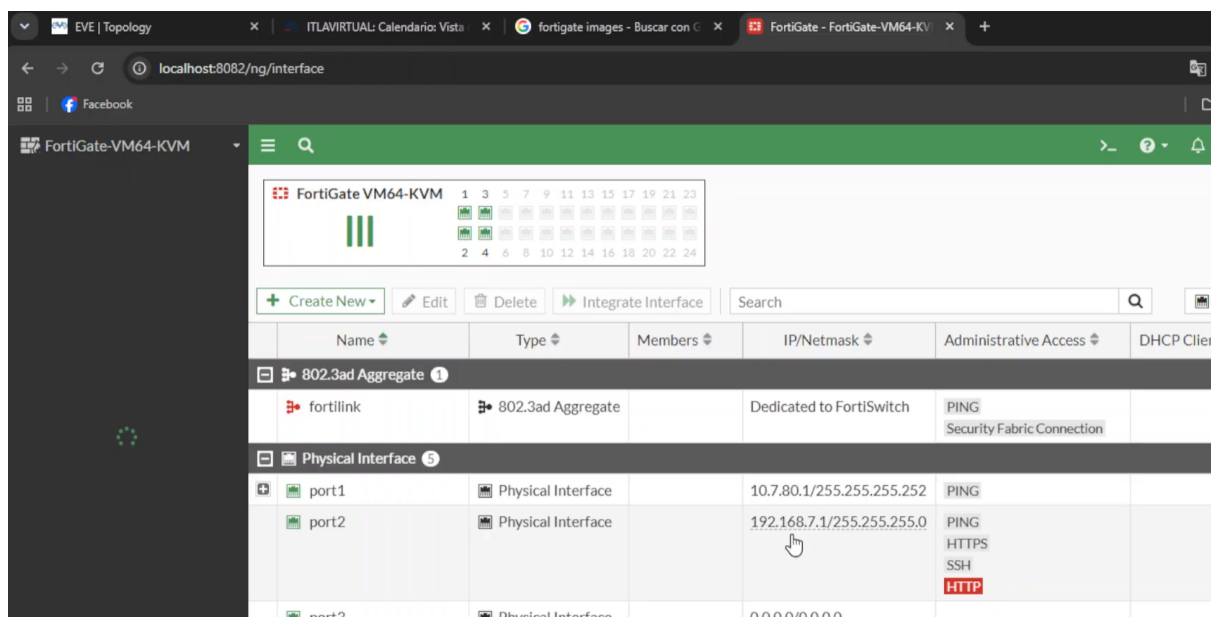
Dispositivo	Interfaz	Dirección IP	Rol
ISP	Fa0/0	10.7.85.1/30	WAN → Linux-Client
ISP	Fa1/0	10.7.80.2/30	WAN → FortiGate
FortiGate	port1	10.7.80.1/30	WAN → ISP
FortiGate	port2	192.168.7.1/24	LAN
Linux-Client	ens3	10.7.85.2/30	WAN (cliente remoto)
Linux-Client	tun0	172.16.78.1/32	IP asignada por VPN Pool
Linux-Server	ens3	192.168.7.2/24	Host LAN

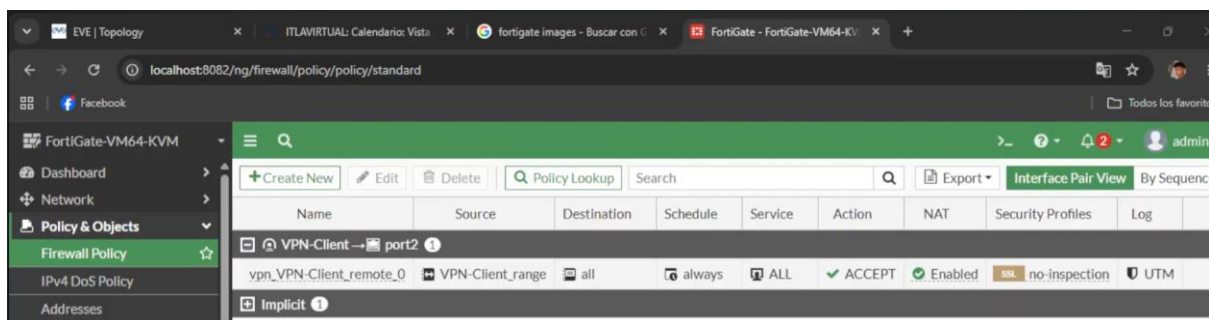
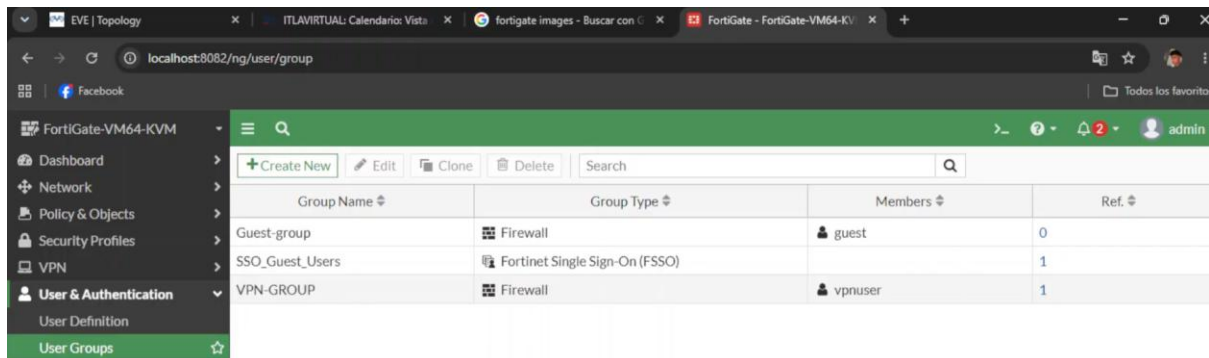
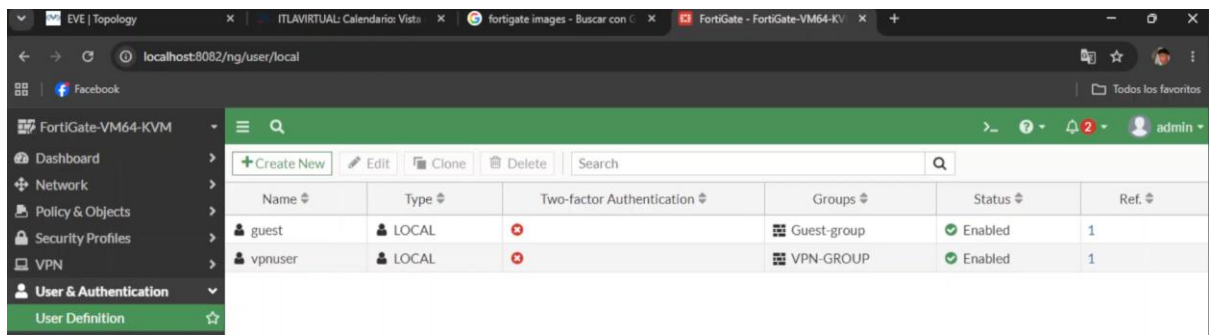
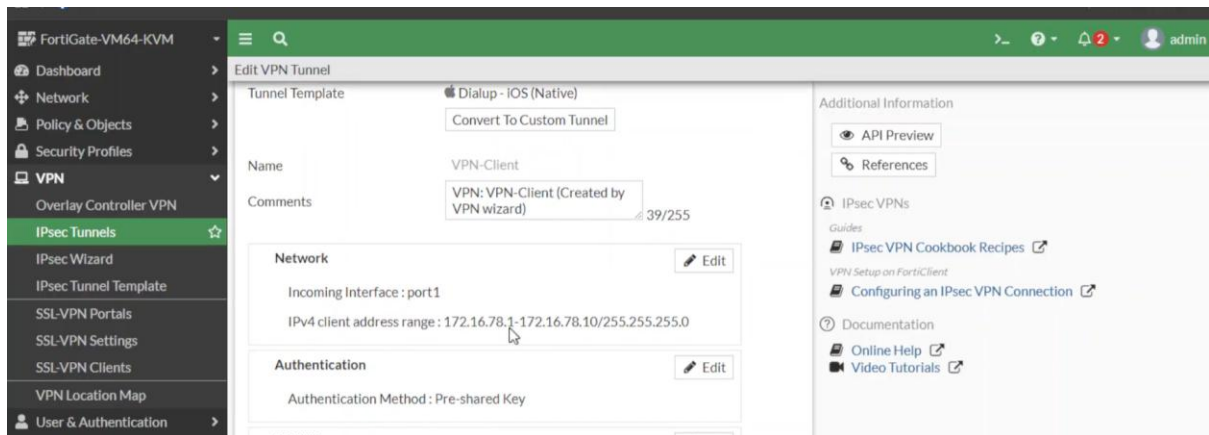
Pool VPN	Inicio	Fin
VPN-Client_range	172.16.78.1	172.16.78.10

2. Parámetros VPN

Parámetro	Valor
Tipo de VPN	IPSec Client-to-Site (Remote Access)
Servidor VPN	FortiGate 7.0.13
Cliente VPN	Ubuntu 22.04 con strongSwan 5.9.5
Plantilla Wizard	Remote Access — Native (iOS)
Fase 1 — Cifrado	DES-CBC
Fase 1 — Hash	MD5
Fase 1 — Autenticación	XAuth + Pre-Shared Key
Fase 1 — Grupo DH	MODP2048
Fase 2 — Cifrado	DES-MD5
Pre-Shared Key	Yeury0780
Usuario XAuth	vpnuser
Contraseña XAuth	Yeury0780
Pool de IPs VPN	172.16.78.1 — 172.16.78.10
Asignación IP	Mode Config

4. Verificación y Funcionamiento





```

traceroute 192.168.7.2
PING 192.168.7.2 (192.168.7.2) 56(84) bytes of data.
64 bytes from 192.168.7.2: icmp_seq=1 ttl=63 time=19.0 ms
64 bytes from 192.168.7.2: icmp_seq=2 ttl=63 time=15.3 ms
64 bytes from 192.168.7.2: icmp_seq=3 ttl=63 time=22.3 ms
64 bytes from 192.168.7.2: icmp_seq=4 ttl=63 time=19.8 ms

--- 192.168.7.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 15.321/19.104/22.284/2.493 ms
traceroute to 192.168.7.2 (192.168.7.2), 30 hops max, 60 byte packets
 1  * * *
 2  192.168.7.2 (192.168.7.2)  26.208 ms  26.264 ms  26.228 ms

```

```

Starting strongSwan 5.9.5 IPsec [starter]...
root@ubuntu:~#
root@ubuntu:~# ipsec up VPN-CLIENT
generating QUICK_MODE request 829906040 [ HASH SA No ID ID ]
sending packet: from 10.7.85.2[500] to 10.7.80.1[500] (164 bytes)
received packet: from 10.7.80.1[500] to 10.7.85.2[500] (148 bytes)
parsed QUICK_MODE response 829906040 [ HASH SA No ID ID ]
selected proposal: ESP:DES_CBC/HMAC_MD5_96/NO_EXT_SEQ
detected rekeying of CHILD_SA VPN-CLIENT{1}
CHILD_SA VPN-CLIENT{2} established with SPIs c7f48ef8_i db3c827b_o and TS 172.16.78.1/32 === 192.168.7.0/24
generating QUICK_MODE request 829906040 [ HASH ]
sending packet: from 10.7.85.2[500] to 10.7.80.1[500] (52 bytes)
connection 'VPN-CLIENT' established successfully
root@ubuntu:~#

```

```

root@ubuntu:~#
root@ubuntu:~# ipsec status
Security Associations (1 up, 0 connecting):
  VPN-CLIENT[1]: ESTABLISHED 25 seconds ago, 10.7.85.2[vpnuser]...10.7.80.1[10.7.80.1]
  VPN-CLIENT{1}: REKEYED, TUNNEL, reqid 1, expires in 11 hours
  VPN-CLIENT{1}: 172.16.78.1/32 === 192.168.7.0/24
  VPN-CLIENT{2}: INSTALLED, TUNNEL, reqid 1, ESP SPIs: c7f48ef8_i db3c827b_o
  VPN-CLIENT{2}: 172.16.78.1/32 === 192.168.7.0/24
root@ubuntu:~#

```

```

2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:50:00:00:01:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.7.85.2/30 scope global ens3
        valid_lft forever preferred_lft forever
    inet 172.16.78.1/32 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::250:ff:fe00:100/64 scope link
        valid_lft forever preferred_lft forever

```